

List the ordered pairs in the relation R from $A = \{0, 1, 2, 3, 4\}$ to $B = \{0, 1, 2, 3\}$, where $(a, b) \in R$ if and only if

a) $a = b$.

b) $a + b = 4$.

c) $a > b$.

d) $a \mid b$.

e) $\gcd(a, b) = 1$.

f) $\text{lcm}(a, b) = 2$.

For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is irreflexive, whether it is symmetric, whether it is antisymmetric, whether it is asymmetric, and whether it is transitive.

a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$

b) $\{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

c) $\{(2, 4), (4, 2)\}$

d) $\{(1, 2), (2, 3), (3, 4)\}$

e) $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$

f) $\{(1, 3), (1, 4), (2, 3), (2, 4), (3, 1), (3, 4)\}$

Determine whether the relation R on the set of all integers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

a) $x \neq y$.

b) $xy \geq 1$.

c) $x = y + 1$ or $x = y - 1$.

d) $x \equiv y \pmod{7}$.

e) x is a multiple of y .

f) x and y are both negative or both nonnegative.

g) $x = y^2$.

h) $x \geq y^2$.

Give an example of a relation on a set that is

a) both symmetric and antisymmetric.

b) neither symmetric nor antisymmetric.

How many different relations are there from a set with m elements to a set with n elements?

Let R be the relation $R = \{(a, b) \mid a \text{ divides } b\}$ on the set of positive integers. Find

a) R^{-1} .

b) $\bar{R}$.

Suppose that the function f from A to B is a one-to-one correspondence. Let R be the relation that equals the graph of f . That is, $R = \{(a, f(a)) \mid a \in A\}$. What is the inverse relation R^{-1} ?

Consider following relations:

$R_1 = \{(a, b) \in \mathbf{R}^2 \mid a > b\}$, the “greater than” relation,

$R_2 = \{(a, b) \in \mathbf{R}^2 \mid a \geq b\}$, the “greater than or equal to” relation,

$R_3 = \{(a, b) \in \mathbf{R}^2 \mid a < b\}$, the “less than” relation,

$R_4 = \{(a, b) \in \mathbf{R}^2 \mid a \leq b\}$, the “less than or equal to” relation,

$R_5 = \{(a, b) \in \mathbf{R}^2 \mid a = b\}$, the “equal to” relation,

$R_6 = \{(a, b) \in \mathbb{R}^2 \mid a \neq b\}$, the “unequal to” relation.

Find

a) $R_2 \cup R_4$.

b) $R_3 \cup R_6$.

c) $R_3 \cap R_6$.

d) $R_4 \cap R_6$.

e) $R_3 - R_6$.

f) $R_6 - R_3$.

g) $R_2 \oplus R_6$.

h) $R_3 \oplus R_5$.

How many relations are there on the set $\{a, b, c, d\}$?

How many relations are there on the set $\{a, b, c, d\}$ that contain the pair (a, a) ?

Find the error in the “proof” of the following “theorem.”

“Theorem”: Let R be a relation on a set A that is symmetric and transitive. Then R is reflexive.

“Proof”: Let $a \in A$. Take an element $b \in A$ such that $(a, b) \in R$. Because R is symmetric, we also have $(b, a) \in R$. Now using the transitive property, we can conclude that $(a, a) \in R$ because $(a, b) \in R$ and $(b, a) \in R$.

Represent each of these relations on $\{1, 2, 3\}$ with a matrix (with the elements of this set listed in increasing order).

a) $\{(1, 1), (1, 2), (1, 3)\}$

b) $\{(1, 2), (2, 1), (2, 2), (3, 3)\}$

c) $\{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

d) $\{(1, 3), (3, 1)\}$

List the ordered pairs in the relations on $\{1, 2, 3\}$ corresponding to these matrices (where the rows and columns correspond to the integers listed in increasing order).

a)
$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

b)
$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

c)
$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

How can the matrix representing a relation R on a set A be used to determine whether the relation is irreflexive?

Determine whether the relations represented by the following matrices are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.

a) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3, \dots, 100\}$ consisting of the first 100 positive integers have if R is

- a) $\{(a, b) \mid a > b\}$?
- b) $\{(a, b) \mid a \neq b\}$?
- c) $\{(a, b) \mid a = b + 1\}$?
- d) $\{(a, b) \mid a = 1\}$?
- e) $\{(a, b) \mid ab = 1\}$?

How can the matrix for $\bar{R}$, the complement of the relation R , be found from the matrix representing R , when R is a relation on a finite set A ?

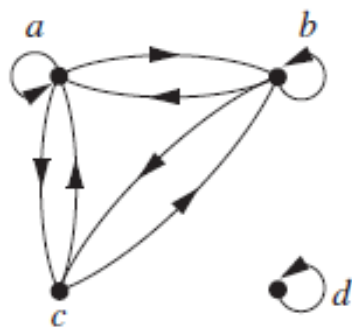
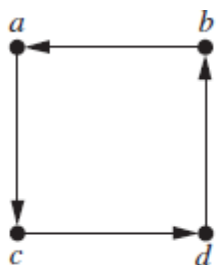
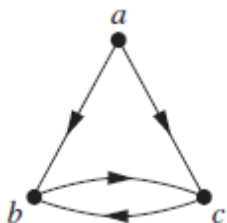
Let R be the relation represented by the matrix

$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Find the matrix representing

- a) R^{-1} .
- b) $\bar{R}$.

List the ordered pairs in the relations represented by the directed graphs.



Determine whether the relations represented by the above directed graphs are reflexive, irreflexive, symmetric, antisymmetric, and/or transitive.

Let R be a relation on a set A . Explain how to use the directed graph representing R to obtain the directed graph representing the inverse relation R^{-1} .

Let R be a relation on a set A . Explain how to use the directed graph representing R to obtain the directed graph representing the complement relation $\bar{R}$.

List the triples in the relation $\{(a, b, c) \mid a, b, \text{ and } c \text{ are integers with } 0 < a < b < c < 5\}$.

Which 4-tuples are in the relation $\{(a, b, c, d) \mid a, b, c, \text{ and } d \text{ are positive integers with } abcd = 6\}$?

List the 5-tuples in the relation in (following) Table 8.

TABLE 8 Flights.				
<i>Airline</i>	<i>Flight_number</i>	<i>Gate</i>	<i>Destination</i>	<i>Departure_time</i>
Nadir	122	34	Detroit	08:10
Acme	221	22	Denver	08:17
Acme	122	33	Anchorage	08:22
Acme	323	34	Honolulu	08:30
Nadir	199	13	Detroit	08:47
Acme	222	22	Denver	09:10
Nadir	322	34	Detroit	09:44

Which of these relations on $\{0, 1, 2, 3\}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack.

- a) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$
- b) $\{(0, 0), (0, 2), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$
- c) $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$
- d) $\{(0, 0), (1, 1), (1, 3), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$
- e) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

Show that the relation R consisting of all pairs (x, y) such that x and y are bit strings that agree in their first and third bits is an equivalence relation on the set of all bit strings of length three or more.

Let R be the relation on the set of ordered pairs of positive integers such that $((a, b), (c, d)) \in R$ if and only if $a + d = b + c$. Show that R is an equivalence relation.